Project: Worked examples
Title: Motor start voltage dip and PF correction
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1. Scope and method

Calculates the voltage dip when a three-phase induction motor starts, from the source, cable and locked-rotor impedances. It also gives the capacitor kvar for a target power factor and the parallel-resonance order.

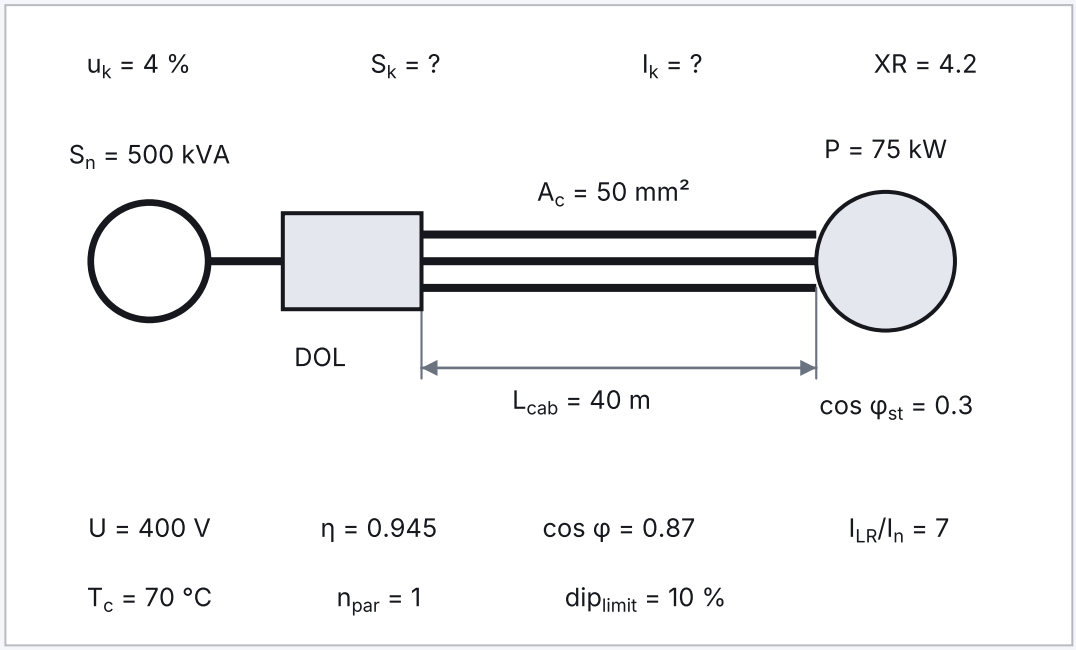


Figure 1 — A motor started direct on line: the source, the contactor, the cable run L_{cab} of section A_c, and the motor P. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Rated current	$I_n = P/(\sqrt{3}\cdot U\cdot\cos\varphi\cdot\eta)$, or entered
Starting current at rated voltage	DOL $I_{LR}\cdot I_n$; star-delta $I_{LR}\cdot I_n/3$; soft starter $\min(k_{soft}\cdot I_n, I_{LR}\cdot I_n)$; drive $k_{vfd}\cdot I_n$
Motor starting impedance (DOL, star-delta, and a soft starter that does not reach its limit)	$\lvert Z_m\rvert = (U/\sqrt{3})/I_{start}$, for a soft starter $(U/\sqrt{3})/(I_{LR}\cdot I_n)$, at the starting power factor, $Z_m = \lvert Z_m\rvert\cdot(\cos\varphi_{st} + j\sin\varphi_{st})$
Regulated current (a drive; a soft starter when its fully conducting current $E/\lvert Z_s + Z_c + Z_{LR}\rvert$ exceeds $k_{soft}\cdot I_n$)	$\lvert I\rvert = I_{start}$ held, lagging V_m by φ; with $R + jX = Z_s + Z_c$, $a = \lvert I\rvert(R\cos\varphi + X\sin\varphi)$, $b = \lvert I\rvert(X\cos\varphi - R\sin\varphi)$: $\lvert V_m\rvert = \sqrt{(E^2 - b^2)} - a$, $V_{pcc} = V_m + I\cdot Z_c$, and $Z_m = V_m/I$ is the equivalent impedance
Source and cable impedance	transformer $\lvert Z_T\rvert = u_k\cdot U^2/S_n$, plus an optional upstream network $\lvert Z_Q\rvert = U^2/S_{kQ}$ in series; fault level $\lvert Z_s\rvert = c\cdot U^2/S_k$ or $c\cdot U/(\sqrt{3}\cdot I_k)$; each element $Z = \lvert Z\rvert\cdot(1 + j\cdot X/R)/\sqrt{(1 + (X/R)^2)}$; cable $R = \rho_T\cdot L/(A\cdot n)$ with $\rho_T = \rho_{20}\cdot[1 + \alpha(T_c - 20)]$, or $R'\cdot L/n$; $X = X'\cdot L/n$

WHAT IT CALCULATES	RELATION
Phasor divider and dip, for a passive motor (phase voltage E = U/√3 at 0°)	$I = E/(Z_s + Z_c + Z_m)$; $V_{pcc} = E\cdot(Z_c + Z_m)/(Z_s + Z_c + Z_m)$; $V_m = E\cdot Z_m/(Z_s + Z_c + Z_m)$; dip = $100\cdot(1 - \lvert V\rvert/E)$ at the motor terminals and at the point of common coupling, checked against the limits; cable drop $dV_{cab} = \sqrt{3}\cdot I\cdot(R\cos\varphi_{st} + X\sin\varphi_{st})$
Power-factor correction	$P_{in} = P/\eta$ (or $\sqrt{3}\cdot U\cdot I_n\cdot\cos\varphi$ when I_n is entered), $Q_c = P_{in}\cdot(\tan\varphi_1 - \tan\varphi_2)$ with φ_1 the motor's (a drive's displacement factor with a drive), $I_c = Q_c/(\sqrt{3}\cdot U)$, I after = $P_{in}/(\sqrt{3}\cdot U\cdot\cos\varphi_2)$
Parallel resonance	$S_{sc} = U^2/\lvert Z_s\rvert$, parallel resonance order $h \approx \sqrt{(S_{sc}/Q_c)}$ ($\lvert Z_s\rvert$ used for X_s, so h reads 1 to 1.4 % low)

Basis: Voltage factor c and the equivalent voltage source[1](cl. 5.3); Limitation of voltage changes and flicker in public low-voltage systems (limits are user inputs)[2](cl. 1); Complex impedance and the potential divider[3](Ch. on AC circuits, 12th ed. (edition to be checked)); Power-factor correction: $Q = P(\tan\varphi_1 - \tan\varphi_2)$[3](Ch. on power factor, 12th ed. (edition to be checked)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
source	Source impedance from	Transformer		drawing: E-SCH-031 rev A	any
$fault_{basis}$	Fault level given as	Fault level S_k (MVA)		default	any
starter	Starter type	Direct on line (DOL)		drawing: E-SCH-031 rev A	any
$cable_{data}$	Cable resistance from	resistivity and area		spec: Cable schedule CS-31: resistivity route as E3	any
harmonics	Harmonic loads present on the supply	no		assumption: No drives or other harmonic loads on the supply.	any
L_{cab}	Cable length, source to motor (links conductor-sizing L)	40	m	drawing: E-LAY-031 rev B	0 … 1000 m
n_{par}	Parallel conductors per phase	1		drawing: E-SCH-031 rev A	1 … 6
A_c	Conductor cross-section (links conductor-sizing CSA)	50	mm²	spec: Cable schedule CS-31: 50 mm² copper	1 … 1000 mm²
ρ_{20}	Resistivity at 20 °C (copper 0.01724, aluminium 0.0282 Ω·mm²/m)	0.01724	Ω·mm²/m	reference: record: annealed copper 1.724e-8 Ω·m at 20 °C, derived physical constant	0.01 … 0.05 Ω·mm²/m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
α	Temperature coefficient of resistance	0.00393	1/K	reference: record: annealed copper, 0.00393 per K at 20 °C, derived physical constant	0.003 … 0.0045 1/K
X_{km}	Reactance per km per conductor	0.08	Ω/km	reference: record: cable reactance 0.08 Ω/km (generic value as the E3 default; replace with the cable datasheet figure)	0 … 1 Ω/km
U	Supply and motor rated voltage (line-to-line)	400	V	spec: Site supply SS-1: 400 V three-phase	50 … 690 V
T_c	Conductor temperature for the resistance	70	°C	assumption: Cable warm from the running load before a restart.	20 … 90 °C
S_k	Three-phase fault level at the point of supply (use the minimum fault level, which gives the deepest dip)	10	MVA	default	0.5 … 500 MVA
I_k	Three-phase fault current at the point of supply (links fault-protection I_{k3})	15	kA	default	1 … 150 kA
S_n	Transformer rating	500	kVA	spec: Transformer datasheet TR-01: 500 kVA rating plate	25 … 5000 kVA
u_k	Transformer impedance voltage	4	%	reference: record: transformer TR-01 impedance voltage 4 % (rating plate; test value invented for this example)	1 … 12 %
XR	Source X/R ratio (of the transformer, or of the fault level)	4.2		reference: record: transformer TR-01 X/R 4.2 (from the load-loss figure; test value invented for this example)	0.5 … 40
XR_Q	Upstream network X/R ratio	10		default	1 … 50

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
k_{soft}	Soft-starter current limit as a multiple of the rated current	3.5		default	1 … 6
k_{vfd}	Drive line-current limit as a multiple of the rated current	1.5		default	0.5 … 2.5
$cos_{phi,vfd}$	Drive line displacement power factor while starting	0.95		default	0.5 … 1
dip_{limit}	Dip limit at the motor terminals	10	%	spec: Machine specification MS-3: motor terminal voltage not below 90 % while starting	1 … 30 %
$cos_{phi,target}$	Target power factor after correction	0.95		spec: Connection agreement CA-2: power factor not below 0.95	0.8 … 1
$dip_{limit,pcc}$	Dip limit at the point of common coupling (when the network operator sets one)	5	%	spec: Connection agreement CA-2: 5 % at the point of common coupling (test value invented for this example)	0.5 … 20 %
P	Motor rated output (shaft) power, from the nameplate (motor-sizing P_{cont} is only a lower bound)	75	kW	spec: Pump schedule PS-03: 75 kW motor rated output (test value invented for this example)	0.1 … 2000 kW
η	Motor efficiency at rated load	0.945		spec: Motor datasheet MD-12: 94.5 % at full load (test value invented for this example)	0.5 … 0.99
cos_{phi}	Motor power factor at rated load	0.87		spec: Motor datasheet MD-12: 0.87 at full load (test value invented for this example)	0.5 … 1
$I_{LR,r}$	Locked-rotor current ratio I_{LR}/I_n	7		reference: record: motor MD-12 locked-rotor current ratio 7.0 (datasheet; test value invented for this example)	3 … 10

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$\cos_{phi,st}$	Power factor at locked rotor (starting)	0.3		reference: record: motor MD-12 locked-rotor power factor 0.3 (datasheet; test value invented for this example)	0.1 … 0.8

3. Assumptions

A1. The supply is a stiff source behind the stated impedance, with voltage E = U/√3 before the start. (Stated limitation.)
A2. A direct-on-line or star-delta start draws current through a passive winding. (Stated limitation.)
A3. A soft starter that does not reach its current limit at the dipped voltage (full conduction) does the same. (Stated limitation.)
A4. The passive winding is a constant impedance equal to the locked-rotor impedance at rated voltage. (Stated limitation.)
A5. The impedance is taken at the starting power factor, so the current falls with the dipped voltage. (Stated limitation.)
A6. A soft starter that reaches its limit and a drive regulate the current. (Regulated-current starters.)
A7. They hold $|I| = I_{start}$ whatever the voltage, so the dip is deeper than the passive model gives. (Regulated-current starters.)
A8. The current phasor lags the voltage at the starter or drive input by the stated power factor. (Regulated-current starters.)
A9. The dip is the steady-state phasor value at the instant of locked rotor. (Stated limitation.)
A10. The current falls as the motor accelerates, so the dip is the worst case. (Stated limitation.)
A11. No other load is running on the supply. (Stated limitation.)
A12. No run-up dynamics, flicker (Pst) or tap-changer action is assessed. (Stated limitation.)
A13. A star-delta start draws one third of the direct-on-line current at rated voltage. (Star-delta connection ratio.)
A14. A star-delta start also gives one third of the starting torque, which is not checked here. (Star-delta connection ratio.)
A15. The current spike at the star-to-delta changeover is not modelled. (Star-delta connection ratio.)
A16. The cable resistance follows the voltage-drop method: from the resistivity, area and temperature, or from R′ per kilometre. (User or record values.)
A17. X′ per kilometre is a user or record value. (User or record values.)
A18. No cable table is embedded. (User or record values.)
A19. The capacitor kvar corrects the running power factor of the motor input power P/η at rated load. (Standard power-factor-correction relation.)
A20. The capacitor sits at the point of common coupling. (Standard power-factor-correction relation.)
A21. The resonance order uses the short-circuit apparent power $U^2/|Z_s|$ at that point. (Standard power-factor-correction relation.)
A22. Using $|Z_s|$ for X_s makes h read 1 to 1.4 % low. (Standard power-factor-correction relation.)
A23. Capacitor switching, detuning and self-excitation on disconnection are not assessed. (Standard power-factor-correction relation.)
A24. The fault-protection loop quantities Z_s and I_f are not the positive-sequence source impedance, so they are not linked. (Linking decision.)
A25. The fault level links the fault-protection I_{k3}. (Linking decision.)
A26. Fault-protection computes I_{k3} with c = 1.1 and $U_n = \sqrt{3}\cdot U_0$ (398.4 V at 230 V). (Linking decision.)
A27. So the I_k basis defaults c to 1.1, and U should be $\sqrt{3}\cdot U_0$ for an exact match. (Linking decision.)
A28. Load-budget declares no impedance port, so the transformer rating and impedance are entered. (Linking decision.)

A29. The source impedance should be taken at the minimum fault level, which gives the deepest dip. (Stated limitation.)
A30. A transformer alone sits on an infinite bus, so it understates the dip and overstates the resonance order. (Stated limitation.)
A31. Entering the upstream network fault level S_{kQ} removes this understatement. (Stated limitation.)
A32. With a drive, the power-factor correction starts from the drive's displacement factor, not the motor's. (Regulated-current starters.)
A33. The drive is itself a harmonic load. (Regulated-current starters.)
A34. With a regulated current, the voltage V_m is at the starter or drive input, not at the motor terminals. (Regulated-current starters.)
A35. The torque factor k_T is reported for that voltage only. (Regulated-current starters.)
A36. harmonics (Harmonic loads present on the supply) is taken as no. (No drives or other harmonic loads on the supply.)
A37. T_c (Conductor temperature for the resistance) is taken as 70 °C. (Cable warm from the running load before a restart.)

4. Working

[10] $I_{rated} = P / (\sqrt{3} * U * \cos_{phi} * \eta)$
$= 75\ kW / (\sqrt{3} \times 400\ V \times 0.87 \times 0.945)$
$= 131.67\ A$
Motor rated current from the rated output

[13] $I_{start} = I_{LR,r} * I_{rated}$
$= 7 \times 131.67\ A$
$= 921.7\ A$
Starting current at rated voltage (direct on line, locked rotor)

[14] $S_{start} = \sqrt{3} * U * I_{start}$
$= \sqrt{3} \times 400\ V \times 921.7\ A$
$= 638.57\ kVA$
Starting apparent power

[18] $Z_T = u_k * U^2 / S_n$
$= 4\ \% \times (400\ V)^2 / 500\ kVA$
$= 0.0128\ ohm$
Source impedance magnitude from the transformer

[19] $R_{src} = Z_T / \sqrt{1 + XR^2}$
$= 0.0128\ ohm / \sqrt{1 + 4.2^2}$
$= 0.0029647\ ohm$
Source resistance

[20] $X_{src} = R_{src} * XR$
$= 0.0029647\ ohm \times 4.2$
$= 0.012452\ ohm$
Source reactance

[21] $Z_{src} = hypot(R_{src}, X_{src})$
$= hypot(0.0029647\ ohm, 0.012452\ ohm)$
$= 0.0128\ ohm$
Source impedance magnitude

[29] $\rho_T = \rho_{20} * (1 + \alpha * (T_c - 20\ degC))$
$= 0.01724\ ohm \cdot mm^2/m \times (1 + 0.00393\ 1/K \times (70\ degC - 20\ degC))$
$= 0.020628\ ohm \cdot mm^2/m$
Resistivity at the operating temperature

[30] $R_{cab} = \rho_T * L_{cab} / (A_c * n_{par})$
$= 0.020628\ ohm \cdot mm^2/m \times 40\ m / (50\ mm^2 \times 1)$
$= 0.016502\ ohm$
Cable resistance (parallel paths combined)

[31] $X_{cab} = X_{km} * L_{cab} / n_{par}$
= 0.08 ohm/km × 40 m / 1
= 0.0032 ohm
Cable reactance
[32] $Z_m = U / (\sqrt{3} * I_{start})$
= 400 V / (sqrt(3) × 921.7 A)
= 0.25056 ohm
Motor starting impedance magnitude
[33] $R_m = Z_m * \cos_{phi,st}$
= 0.25056 ohm × 0.3
= 0.075168 ohm
Motor starting resistance
[34] $X_m = Z_m * \sqrt{1 - \cos_{phi,st}^2}$
= 0.25056 ohm × sqrt(1 - 0.3^2)
= 0.23902 ohm
Motor starting reactance
[35] $E_{ph} = E = U/\sqrt{3}$ (reference phasor, angle 0°)
= 400 V / √3
= 230.94 V
Source phase voltage before the start
[36] $Z_{s,ph} = Z_s = R_s + jX_s$
= R_s = 0.0029647 Ω, X_s = 0.012452 Ω
= "0.0029647 + j0.012452 Ω [0.0128 Ω ∠ 76.608°]"
Source impedance phasor
[37] $Z_{c,ph} = Z_c = R_c + jX_c$
= R_c = 0.016502 Ω, X_c = 0.0032 Ω
= "0.016502 + j0.0032 Ω [0.01681 Ω ∠ 10.974°]"
Cable impedance phasor
[38] $Z_{m,ph} = Z_m = |Z_m| \cdot (\cos \varphi_{st} + j \sin \varphi_{st})$
= 0.25056 Ω · (0.3 + j0.95394)
= "0.075168 + j0.23902 Ω [0.25056 Ω ∠ 72.542°]"
Motor starting impedance phasor
[39] $Z_{t,ph} = Z_t = Z_s + Z_c + Z_m$
= (0.0029647 + j0.012452) + (0.016502 + j0.0032) + (0.075168 + j0.23902)
= "0.094635 + j0.25467 Ω [0.27169 Ω ∠ 69.615°]"
Total series impedance phasor
[40] $I_{ph} = I = E/Z_t$
= 230.94 V / (0.27169 Ω ∠ 69.615°)
= "296.09 - j796.79 A [850.03 A ∠ -69.615°]"
Starting current phasor at the dipped voltage
[41] $V_{m,ph} = V_m = E \cdot Z_m/Z_t$
= 230.94 V · (0.25056 Ω ∠ 72.542°) / (0.27169 Ω ∠ 69.615°)
= "212.7 + j10.877 V [212.98 V ∠ 2.9273°]"
Motor terminal voltage phasor (phase)
[42] $V_{pcc,ph} = V_{pcc} = E \cdot (Z_c + Z_m)/Z_t$
= 230.94 V · (0.25899 Ω ∠ 69.27°) / (0.27169 Ω ∠ 69.615°)
= "220.14 - j1.3245 V [220.14 V ∠ -0.34473°]"
Point of common coupling voltage phasor (phase)
[43] $I_{dip} = |I|$
= 230.94 V / 0.27169 Ω
= 850.03 A
Starting current at the dipped voltage

[44] $V_m = |V_m| \cdot \sqrt{3}$
= 212.98 V · √3
= 368.9 V
Voltage at the motor terminals while starting (line)
[45] $V_{pcc} = |V_{pcc}| \cdot \sqrt{3}$
= 220.14 V · √3
= 381.3 V
Point of common coupling voltage while starting (line)
[46] $start_{dip} = 100 \cdot (1 - |V_m|/E)$
= 100 · (1 − 212.98 / 230.94)
= 7.7757 %
Voltage dip at the motor terminals
[47] $dip_{pcc} = 100 \cdot (1 - |V_{pcc}|/E)$
= 100 · (1 − 220.14 / 230.94)
= 4.6746 %
Voltage dip at the point of common coupling
[48] $dV_{cab} = \sqrt{3} \cdot Re[I \cdot Z_c \cdot \bar{V}_m/|V_m|]$
= √3 · Re[(850.03 A ∠ −69.615°) · (0.01681 Ω ∠ 10.974°) · e^(−j2.9273°)]
= 11.783 V
Cable drop along the starting voltage: √3·I·(R cos φ + X sin φ), the E3 three-phase form at this current
[49] $k_T = (|V_m|/E)^2$
= (212.98 / 230.94)²
= 0.85053
Torque factor at the dipped voltage relative to rated voltage (torque varies as V²)
[50] $u_{dip} = start_{dip} / dip_{limit}$
= 7.7757 % / 10 %
= 0.77757
Dip at the motor terminals against the limit
[51] $P_{in} = P / \eta$
= 75 kW / 0.945
= 79.365 kW
Motor input power at rated load
[52] $tan_1 = sqrt(1 - cos_{phi}^2) / cos_{phi}$
= sqrt(1 − 0.87^2) / 0.87
= 0.56673
tan φ_1 at the running power factor
[53] $tan_2 = sqrt(1 - cos_{phi,target}^2) / cos_{phi,target}$
= sqrt(1 − 0.95^2) / 0.95
= 0.32868
tan φ_2 at the target power factor
[54] $Q_c = max(0\ kW, P_{in} * (tan_1 - tan_2))$
= max(0 kW, 79.365 kW × (0.56673 − 0.32868))
= 18.892 kvar
Capacitor kvar to reach the target power factor
[55] $I_{cap} = Q_c / (sqrt(3) * U)$
= 18.892 kvar / (sqrt(3) × 400 V)
= 27.269 A
Capacitor line current at rated voltage
[56] $I_{run1} = P_{in} / (sqrt(3) * U * cos_{phi})$
= 79.365 kW / (sqrt(3) × 400 V × 0.87)
= 131.67 A
Running line current before correction

[57] I_{run2} = P_{in} / (sqrt(3) * U * $cos_{phi,target}$)
= 79.365 kW / (sqrt(3) × 400 V × 0.95)
= 120.58 A
Running line current after correction (when the target is reached)
[58] S_{sc} = U^2 / Z_{src}
= (400 V)^2 / 0.0128 ohm
= 12.5 MVA
Short-circuit apparent power at the point of supply
[59] h_{res} = sqrt(S_{sc} / Q_c)
= sqrt(12.5 MVA / 18.892 kvar)
= 25.723
Parallel resonance order of the capacitor bank with the supply
[61] $u_{dip,pcc}$ = dip_{pcc} / $dip_{limit,pcc}$
= 4.6746 % / 5 %
= 0.93491
Point-of-common-coupling dip against the limit

5. Results

SYMBOL	VALUE	DESCRIPTION
$start_{dip}$	7.776 %	Voltage dip at the motor terminals
kvar	18.89 kvar	Capacitor kvar to the target power factor
I_{start}	921.70 A	Starting current at rated voltage
I_{rated}	131.67 A	Motor rated current
Z_{src}	0.01280 Ω	Source impedance (magnitude)
Z_m	0.2506 Ω	Motor starting impedance (magnitude)
S_{start}	638.57 kVA	Starting apparent power
I_{dip}	850.03 A	Starting current at the dipped voltage
V_m	368.9 V	Voltage at the motor terminals while starting
V_{pcc}	381.3 V	Point of common coupling voltage while starting
dip_{pcc}	4.675 %	Voltage dip at the point of common coupling (governing)
dV_{cab}	11.78 V	Cable drop along the starting voltage
k_T	0.85053	Torque factor at the dipped voltage
P_{in}	79.37 kW	Motor input power at rated load
I_{cap}	27.27 A	Capacitor current
I_{run1}	131.67 A	Running current before correction
I_{run2}	120.58 A	Running current after correction
S_{sc}	12.500 MVA	Short-circuit apparent power at the bank
h_{res}	25.723	Parallel resonance order
u_{dip}	0.78	Dip utilisation at the motor terminals
$u_{dip,pcc}$	0.93	Point-of-common-coupling dip utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Point-of-common-coupling dip against the limit	4.675 %	5 %	0.93	[2](cl. 1)
PASS	Motor-terminal dip against the limit	7.776 %	10 %	0.78	

7. Applicability

WARNING records: tabulated values not taken from a reference record are X_{km} (Cable impedance (resistance and reactance per kilometre)); ρ_{20}, α (Conductor material constants); $I_{LR,r}$, $\cos_{phi,st}$ (Motor starting data); u_k, XR (Transformer impedance data). Enter each as one so its source, edition and licence travel with it. [record]

Defaults used (inputs left empty):
$fault_{basis}$: not entered, so the default of 'Fault level S_k (MVA)' is used.
S_k: not entered, so the default of 10 MVA is used.
I_k: not entered, so the default of 15 kA is used.
XR_Q: not entered, so the default of 10 is used.
k_{soft}: not entered, so the default of 3.5 is used.
k_{vfd}: not entered, so the default of 1.5 is used.
$\cos_{phi,vfd}$: not entered, so the default of 0.95 is used.

8. Verification

```
Template elec.motor-start@1.3.0 Content hash da767ce16b10e18b Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. IEC 60909-0:2016.
2. IEC 61000-3-11:2017.
3. Hughes E. Hughes Electrical and Electronic Technology. 12th ed. Pearson; 2016.